Block practice · The quadratic function

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Completing the square

Reading the maximum or minimum from completed-square form

Sum and product of roots

The discriminant: equal, real or no real roots

Forming a new quadratic from the roots of another

OLevels.com SCHOOL

Practice workbook · Completing the square

Rewriting ax^2 + bx + c in the form a(x + p)^2 + q, or as q - a(x + p)^2 when the x squared term is negative, and stating the constants.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-03

1. Question 3 · part a · 3 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part a only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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1. Question 3 · part a · 3 marks · Working and answer

Working and answer

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2. Question 10 · part a · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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2. Question 10 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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P 6 2 2 8 0 A 0 1 4 3 2

3. Question 7 · part a · 2 marks · Working and answer

Working and answer

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4. Question 2 · part a · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

2 $f(x) = 2x^2 - 12x + 5$

Given that $f(x)$ can be written in the form $f(x) = a(x + b)^2 + c$ where a, b and c are integers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence find the set of values of x for which $f(x) - 37 > 0$

(4)

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P 6 6 3 1 0 A 0 4 3 6

4. Question 2 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part a · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

2 $f(x) = 2x^2 + 4x + 9$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$, where A, B and C are integers,

(a) find the value of A, the value of B and the value of C

(3)

(b) Hence, or otherwise, find

(i) the value of x for which $\frac{1}{f(x)}$ is a maximum

(ii) the maximum value of $\frac{1}{f(x)}$

(2)

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5. Question 2 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part a · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 $f(x) = 7 + 4x - x^2$

(a) Write $f(x)$ in the form $a - b(x + c)^2$ where a, b and c are integers to be found. (3)

(b) Hence, or otherwise, find

(i) the maximum value of $f(x)$

(ii) the value of x for which this maximum occurs. (2)

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P 6 0 4 7 5 A 0 4 3 6

6. Question 2 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 2 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

2 $f(x) = 2x^2 - 10x + 7$

Given that $f(x)$ can be written in the form $f(x) = a(x+b)^2 + c$ where a, b and c are rational numbers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence, or otherwise, write down

(i) the minimum value of $f(x)$

(ii) the value of x at which this minimum value occurs.

(2)

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7. Question 2 · part a · 3 marks · Working and answer

Working and answer

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8. Question 2 · part a · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

2 $f(x) = 2 + \frac{4}{5}x - \frac{1}{25}x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(4)

(b) Hence write down

(i) the maximum value of $f(x)$,

(ii) the value of x for which this maximum occurs.

(2)

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P 6 6 0 2 5 A 0 4 3 6

8. Question 2 · part a · 4 marks · Working and answer

Working and answer

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9. Question 11 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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30

P 7 2 8 6 4 A 0 3 0 3 2

9. Question 11 · part a · 4 marks · Working and answer

Working and answer

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10. Question 10 · part a · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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10. Question 10 · part a · 4 marks · Working and answer

Working and answer

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11. Question 2 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

2 (a) Express $x^2 + 4x - 8$ in the form $(x + a)^2 + b$ where a and b are constants whose values are to be found.

(2)

(b) Use algebra to solve the simultaneous equations

$$y = x^2 + 4x - 8$$
$$y = 2x + 7$$

(5)

The curve C has equation $y = x^2 + 4x - 8$
The straight line L has equation $y = 2x + 7$

Using the same axes and the results of part (a) and part (b),

(c) sketch C and L, showing clearly the coordinates of the turning point of C and the coordinates of the points of intersection of C and L.

(4)

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11. Question 2 · part a · 2 marks · Working and answer

Working and answer

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12. Question 2 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

2 $g(x) = 2x^2 + \frac{1}{2}x - 3$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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12. Question 2 · part a · 3 marks · Working and answer

Working and answer

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Practice workbook · Reading the maximum or minimum from completed-square form

Reading the maximum or minimum value of a quadratic, and the x-value where it happens, from its completed-square form, including simple related functions.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-04

1. Question 7 · part b.ii · 2 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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1. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 10 · part b.i · 0 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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2. Question 10 · part b.i · 0 marks · Working and answer

Working and answer

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3. Question 10 · part b.ii · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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3. Question 10 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 7 · part b.i · 0 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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4. Question 7 · part b.i · 0 marks · Working and answer

Working and answer

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5. Question 3 · part b.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part b.i only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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5. Question 3 · part b.i · 0 marks · Working and answer

Working and answer

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6. Question 3 · part b.ii · 2 marks

2023 JAN · Paper 02

Further practice · Attempt part b.ii only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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6. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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7. Question 2 · part b.i · 0 marks

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

2 $f(x) = 2x^2 + 4x + 9$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$, where A, B and C are integers,

(a) find the value of A, the value of B and the value of C

(3)

(b) Hence, or otherwise, find

(i) the value of x for which $\frac{1}{f(x)}$ is a maximum

(ii) the maximum value of $\frac{1}{f(x)}$

(2)

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7. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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8. Question 2 · part b.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

2 $f(x) = 2x^2 + 4x + 9$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$, where A, B and C are integers,

(a) find the value of A, the value of B and the value of C

(3)

(b) Hence, or otherwise, find

(i) the value of x for which $\frac{1}{f(x)}$ is a maximum

(ii) the maximum value of $\frac{1}{f(x)}$

(2)

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8. Question 2 · part b.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

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Question 2 continued

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(Total for Question 2 is 5 marks)

8. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 2 · part b.i · 0 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

2 $f(x) = 7 + 4x - x^2$

(a) Write $f(x)$ in the form $a - b(x + c)^2$ where a, b and c are integers to be found. **(3)**

(b) Hence, or otherwise, find

(i) the maximum value of $f(x)$

(ii) the value of x for which this maximum occurs. **(2)**

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9. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 2 · part b.ii · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

2 $f(x) = 7 + 4x - x^2$

(a) Write $f(x)$ in the form $a - b(x + c)^2$ where a, b and c are integers to be found. **(3)**

(b) Hence, or otherwise, find

(i) the maximum value of $f(x)$

(ii) the value of x for which this maximum occurs. **(2)**

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10. Question 2 · part b.ii · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

Question 2 continued

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(Total for Question 2 is 5 marks)

10. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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11. Question 2 · part b.i · 0 marks

2025 NOV · Paper 01

Further practice · Attempt part b.i only.

2 $f(x) = 2x^2 - 10x + 7$

Given that $f(x)$ can be written in the form $f(x) = a(x+b)^2 + c$ where a, b and c are rational numbers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence, or otherwise, write down

(i) the minimum value of $f(x)$

(ii) the value of x at which this minimum value occurs.

(2)

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11. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 2 · part b.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

2 $f(x) = 2x^2 - 10x + 7$

Given that $f(x)$ can be written in the form $f(x) = a(x+b)^2 + c$ where a, b and c are rational numbers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence, or otherwise, write down

(i) the minimum value of $f(x)$

(ii) the value of x at which this minimum value occurs.

(2)

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12. Question 2 · part b.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

Question 2 continued

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(Total for Question 2 is 5 marks)

12. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 2 · part b.i · 0 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

2 $f(x) = 2 + \frac{4}{5}x - \frac{1}{25}x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(4)

(b) Hence write down

(i) the maximum value of $f(x)$,

(ii) the value of x for which this maximum occurs.

(2)

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13. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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14. Question 2 · part b.ii · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 $f(x) = 2 + \frac{4}{5}x - \frac{1}{25}x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(4)

(b) Hence write down

(i) the maximum value of $f(x)$,

(ii) the value of x for which this maximum occurs.

(2)

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14. Question 2 · part b.ii · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.



Question 2 continued

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(Total for Question 2 is 6 marks)

14. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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15. Question 11 · part b.i · 0 marks

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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15. Question 11 · part b.i · 0 marks · Working and answer

Working and answer

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16. Question 11 · part b.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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16. Question 11 · part b.ii · 2 marks · Working and answer

Working and answer

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17. Question 10 · part b · 1 mark

2024 NOV · Paper 01

Further practice · Attempt part b only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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17. Question 10 · part b · 1 mark · Working and answer

Working and answer

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18. Question 2 · part b.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part b.i only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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18. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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19. Question 2 · part b.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part b.ii only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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P 7 3 5 8 4 A 0 6 3 6

19. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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20. Question 2 · part c.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part c.i only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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20. Question 2 · part c.i · 0 marks · Working and answer

Working and answer

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21. Question 2 · part c.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part c.ii only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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21. Question 2 · part c.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part c.ii only.

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Question 2 continued

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(Total for Question 2 is 7 marks)

21. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Sum and product of roots

Writing down alpha + beta and alpha*beta from the coefficients and using them to find expressions such as alpha^2 + beta^2, alpha^3 + beta^3, alpha^4 + beta^4 or fractions in the roots, without solving the equation.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-07

1. Question 11 · part b.i · 2 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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1. Question 11 · part b.i · 2 marks · Working and answer

Working and answer

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2. Question 10 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

10 The roots of a quadratic equation are α and β where

$$\alpha+\beta=-\frac{5}{2} \text{ and } \alpha^3+\beta^3=\frac{115}{8}$$

(a) Show that $\alpha\beta=4$

(3)

(b) Form a quadratic equation with integer coefficients, that has roots

$$\frac{\alpha^2+1}{\beta} \text{ and } \frac{\beta^2+1}{\alpha}$$

(7)

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2. Question 10 · part a · 3 marks · Working and answer

Working and answer

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3. Question 5 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

5 (a) Show that $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=\alpha^3+\beta^3$

(2)

The quadratic equation $2x^2-6x-7=0$ has roots α and β

Without solving the equation

(b) form a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$

(6)

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3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

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4. Question 4 · part a · 4 marks · Working and answer

Working and answer



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5. Question 4 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

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5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 8 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 The quadratic equation $3x^2 - kx - 1 = 0$, where k is a positive integer, has roots α and β

(a) Show that $\alpha^2 + \beta^2 = \dfrac{k^2 + 6}{9}$

(3)

Given that $\alpha^4 + \beta^4 = \dfrac{466}{81}$

(b) find the value of k

(5)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha^3 + \beta}{\beta} \text{ and } \frac{\beta^3 + \alpha}{\alpha}$$

(6)

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6. Question 8 · part a · 3 marks · Working and answer

Working and answer

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7. Question 10 · part a.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part a.i only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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7. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 10 · part a.ii · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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8. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

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9. Question 8 · part b · 4 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

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9. Question 8 · part b · 4 marks · Working and answer

Working and answer

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10. Question 7 · part a · 1 mark

2025 NOV · Paper 01

Further practice · Attempt part a only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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10. Question 7 · part a · 1 mark · Working and answer

Working and answer

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11. Question 7 · part c.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c.i only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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11. Question 7 · part c.i · 3 marks · Working and answer

Working and answer

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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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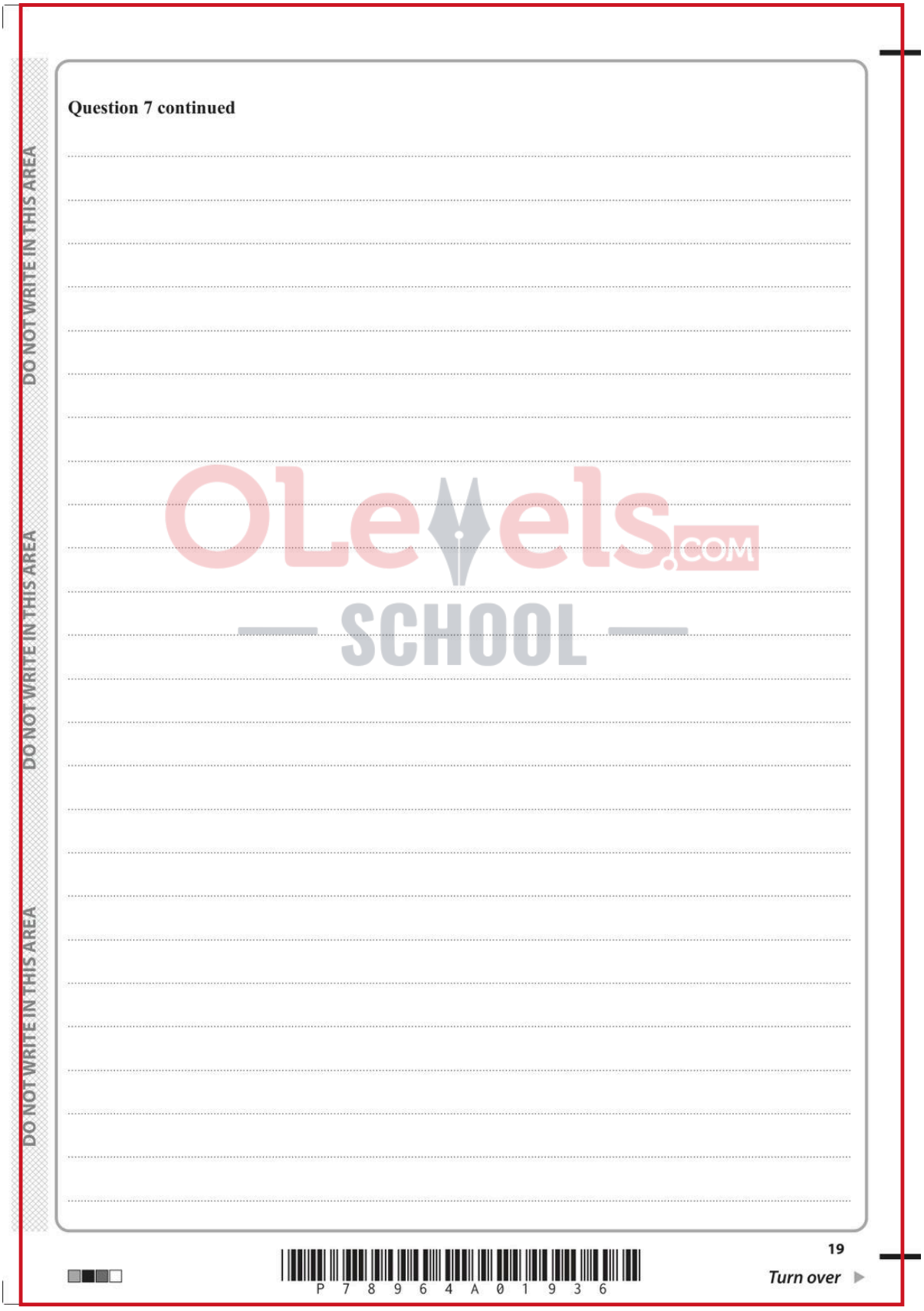
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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.



Question 7 continued

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Turn over

12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

Question 7 continued

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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

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Question 7 continued

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(Total for Question 7 is 10 marks)

12. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

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13. Question 5 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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13. Question 5 · part a · 2 marks · Working and answer

Working and answer

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14. Question 5 · part b · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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14. Question 5 · part b · 2 marks · Working and answer

Working and answer

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15. Question 5 · part c · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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15. Question 5 · part c · 2 marks · Working and answer

Working and answer

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16. Question 10 · part d · 1 mark

2024 NOV · Paper 01

Further practice · Attempt part d only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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16. Question 10 · part d · 1 mark · Working and answer

Working and answer

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17. Question 5 · part a.i · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

5 The roots of the quadratic equation $2x^2 + (6 + 2p)x + 2p = 0$ are α and β

(a) Write down an expression in terms of p for

(i) $\alpha + \beta$ (ii) $\alpha\beta$

(2)

(b) Show that $(\alpha - \beta)^2 = 9 + 2p + p^2$

(4)

Given that $(\alpha - \beta) = 3$

(c) find the possible values of p

(3)

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17. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 8 · part b.i · 0 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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18. Question 8 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 8 · part b.ii · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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19. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

20. Question 11 · part a.i · 0 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

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20. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

21. Question 8 · part a · 1 mark

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

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21. Question 8 · part a · 1 mark · Working and answer

Working and answer

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Practice workbook · The discriminant: equal, real or no real roots

Conditions on the discriminant of a quadratic, either given directly or obtained by substituting a line into a curve or rearranging a curve as a quadratic in x.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-05

1. Question 1 · part whole · 4 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

1 Find the set of values of k for which the equation

$$2kx^2 + 5kx + 5k - 3 = 0 \qquad \text{where } k \neq 0$$

has real roots.

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(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 5 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

1 The equation $kx^2 + 8x + 3k = 0$ where k is a constant, has real unequal roots.

Find the set of values of k giving your answer in an exact simplified form.

(5)

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(Total for Question 1 is 5 marks)

2. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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3. Question 7 · part a · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

7 (a) Find the set of values of k for which the equation $kx^2 - 4x + 2k = 7$ has real roots

(4)

Given that the roots of the equation $kx^2 - 4x + 2k = 7$ are α and β,

(b) form a quadratic equation with roots $\frac{\alpha + 1}{\alpha}$ and $\frac{\beta + 1}{\beta}$

Give each coefficient in terms of k.

(8)

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3. Question 7 · part a · 4 marks · Working and answer

Working and answer

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4. Question 1 · part whole · 6 marks

2021 JUN · Paper 01

Further practice · Attempt part whole only.

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1 The quadratic equation

$$3(k + 2)x^2 + (k + 5)x + k = 0$$

has real roots.

Find the set of possible values of k.

(6)

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(Total for Question 1 is 6 marks)

4. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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5. Question 1 · part whole · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 The quadratic equation

$$kx^2 - (2k + 6)x + 16 = 0$$

has equal roots.

Find the two possible values of k

(4)

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(Total for Question 1 is 4 marks)

5. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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6. Question 2 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

2 $f(x) = 2x^2 + (p - 1)x - 2p$ where p is a constant.

Find the set of values of p for which the equation $f(x) = 0$ has two distinct real roots.

(5)

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6. Question 2 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 5 marks)

6. Question 2 · part whole · 5 marks · Working and answer

Working and answer

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7. Question 1 · part whole · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 $f(x) = 2x^2 + (k+8)x + k$

Show that for all values of k, the equation $f(x) = 0$ has distinct real roots.

(4)

(Total for Question 1 is 4 marks)

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7. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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8. Question 9 · part b · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

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8. Question 9 · part b · 4 marks · Working and answer

Working and answer

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9. Question 9 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

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9. Question 9 · part a · 3 marks · Working and answer

Working and answer

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10. Question 5 · part a · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

5 Given that k is a non-zero constant

curve C has equation $kx^2 - xy + (k+1)x = 1$

straight line l has equation $y = \frac{k}{2}x + 1$

The point A is the only point that lies on both C and l.

(a) Find the value of k

(6)

(b) Hence, find the coordinates of A.

(2)

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10. Question 5 · part a · 6 marks · Working and answer

Working and answer

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11. Question 10 · part c · 9 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

10 The curve C has equation $y = \dfrac{ax - 5}{b - x}$ where a and b are integers and $x \neq b$

One intersection of C with the coordinate axes is at the point with coordinates $\left(\frac{5}{4}, 0\right)$

The asymptote parallel to the y-axis has equation $x = 3$

(a) Find the value of a and the value of b

(2)

(b) Sketch C, showing clearly the asymptotes with their equations and the coordinates of the points of intersection with the coordinate axes.

(5)

The straight line l with equation $4y - 7x = k$ has no points of intersection with C

(c) Show, using algebra, that the range of possible values of k can be written as

$$m < k < n$$

where m and n are integers to be found.

(9)

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11. Question 10 · part c · 9 marks

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Further practice · Attempt part c only.

Question 10 continued

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11. Question 10 · part c · 9 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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11. Question 10 · part c · 9 marks

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Further practice · Attempt part c only.

Question 10 continued

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11. Question 10 · part c · 9 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 16 marks)

11. Question 10 · part c · 9 marks · Working and answer

Working and answer

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 14 marks)

12. Question 11 · part d · 7 marks · Working and answer

Working and answer

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Practice workbook · Forming a new quadratic from the roots of another

Building a quadratic equation, usually with integer coefficients, whose roots are given expressions in alpha and beta, using only the sum and product of the original roots.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-10

1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

6 (a) Show that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

(3)

The quadratic equation $x^2 - 7kx + k^2 = 0$, where k is a positive constant, has roots α and β where $\alpha > \beta$

(b) Show that $\alpha - \beta = 3k\sqrt{5}$

(3)

(c) Hence form a quadratic equation with roots $\alpha + 1$ and $\beta - 1$

Give your equation in the form $x^2 + px + q = 0$ where p and q should be given in terms of k.

(4)

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1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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1. Question 6 · part c · 4 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 10 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

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2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

10 The roots of a quadratic equation are α and β where

$$\alpha+\beta=-\frac{5}{2} \text{ and } \alpha^3+\beta^3=\frac{115}{8}$$

(a) Show that $\alpha\beta=4$

(3)

(b) Form a quadratic equation with integer coefficients, that has roots

$$\frac{\alpha^2+1}{\beta} \text{ and } \frac{\beta^2+1}{\alpha}$$

(7)

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2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 10 marks)

2. Question 10 · part b · 7 marks · Working and answer

Working and answer

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3. Question 5 · part b · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

5 (a) Show that $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=\alpha^3+\beta^3$ (2)

The quadratic equation $2x^2-6x-7=0$ has roots α and β

Without solving the equation

(b) form a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$ (6)

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P 7 2 8 6 5 A 0 1 6 3 6

3. Question 5 · part b · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

3. Question 5 · part b · 6 marks · Working and answer

Working and answer

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4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

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4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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4. Question 11 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 12 marks)

4. Question 11 · part c · 5 marks · Working and answer

Working and answer

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5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

7 (a) Find the set of values of k for which the equation $kx^2 - 4x + 2k = 7$ has real roots

(4)

Given that the roots of the equation $kx^2 - 4x + 2k = 7$ are α and β,

(b) form a quadratic equation with roots $\frac{\alpha + 1}{\alpha}$ and $\frac{\beta + 1}{\beta}$

Give each coefficient in terms of k.

(8)

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5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 12 marks)

5. Question 7 · part b · 8 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

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6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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10

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6. Question 4 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 10 marks)

6. Question 4 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 The quadratic equation $3x^2 - kx - 1 = 0$, where k is a positive integer, has roots α and β

(a) Show that $\alpha^2 + \beta^2 = \dfrac{k^2 + 6}{9}$

(3)

Given that $\alpha^4 + \beta^4 = \dfrac{466}{81}$

(b) find the value of k

(5)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha^3 + \beta}{\beta} \text{ and } \frac{\beta^3 + \alpha}{\alpha}$$

(6)

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7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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7. Question 8 · part c · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 14 marks)

7. Question 8 · part c · 6 marks · Working and answer

Working and answer

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8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 The quadratic equation $2x^2 + kx + 4 = 0$ has roots α and β such that

$$k < 0 \text{ and } \alpha > \beta$$

Given that $\alpha^2 - \beta^2 = \frac{7\sqrt{17}}{4}$

(a) show that $k = -7$

(8)

(b) Hence form a quadratic equation that has roots

$$(\alpha - \beta) \text{ and } (\alpha + \beta)$$

(4)

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8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

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8. Question 10 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 10 continued

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(Total for Question 10 is 12 marks)

8. Question 10 · part b · 4 marks · Working and answer

Working and answer

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9. Question 6 · part a · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

6 $f(x) = 4x^2 - 3x - 5$

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(a) form an equation, with integer coefficients, that has roots $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$ **(6)**

$g(x) = 4x^2 + px + q$ where p and q are constants

Given that the equation $g(x) = 0$ has roots $3\alpha + \beta$ and $\alpha + 3\beta$

(b) find the value of p and the value of q **(5)**

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9. Question 6 · part a · 6 marks · Working and answer

Working and answer

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10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

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10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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10. Question 8 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

10. Question 8 · part c · 5 marks · Working and answer

Working and answer

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11. Question 2 · part whole · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

2 The quadratic equation $3x^2 - 5x + 1 = 0$ has roots α and β

Without solving the equation,

form a quadratic equation with integer coefficients, that has roots $\frac{\alpha}{2\beta}$ and $\frac{\beta}{2\alpha}$ **(8)**

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11. Question 2 · part whole · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 8 marks)

11. Question 2 · part whole · 8 marks · Working and answer

Working and answer

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12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

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12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

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12. Question 5 · part d · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 5 continued

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(Total for Question 5 is 12 marks)

12. Question 5 · part d · 6 marks · Working and answer

Working and answer

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13. Question 1 · part whole · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

1 The roots of the equation $4x^2 - 3x - 8 = 0$ are α and β

Without solving this equation, form a quadratic equation, with integer coefficients, which has roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

(7)

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(Total for Question 1 is 7 marks)

13. Question 1 · part whole · 7 marks · Working and answer

Working and answer

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14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 $f(x) = 2x^2 + 5x - p$

The equation $f(x) = 0$ has roots α and β

Given that $\alpha^3 + \beta^3 = -\frac{215}{8}$

(a) find the value of p

(5)

Without solving the equation $f(x) = 0$

(b) form a quadratic equation, with integer coefficients, that has roots

$$\frac{\alpha + \beta}{\alpha^2} \text{ and } \frac{\alpha + \beta}{\beta^2}$$

(5)

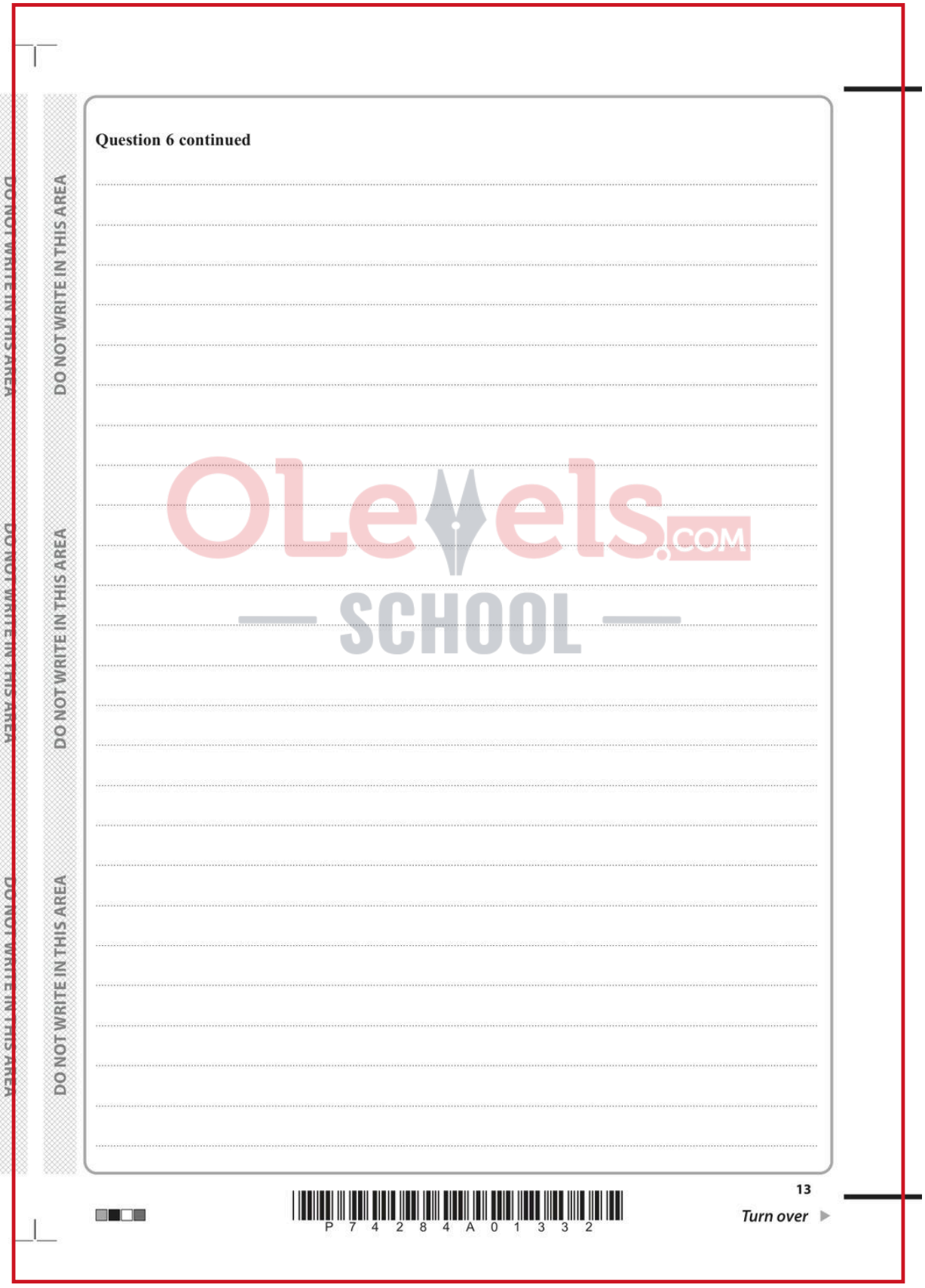
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14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 6 continued

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14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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14. Question 6 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 10 marks)

14. Question 6 · part b · 5 marks · Working and answer

Working and answer

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15. Question 10 · part c · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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15. Question 10 · part c · 6 marks · Working and answer

Working and answer

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16. Question 5 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

5 Given that α and β are such that $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$

(a) form a quadratic equation with integer coefficients that has roots α and β,

(2)

(b) form a quadratic equation with integer coefficients that has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(6)

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16. Question 5 · part a · 2 marks · Working and answer

Working and answer

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17. Question 5 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

5 Given that α and β are such that $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$

(a) form a quadratic equation with integer coefficients that has roots α and β,

(2)

(b) form a quadratic equation with integer coefficients that has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(6)

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17. Question 5 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

17. Question 5 · part b · 6 marks · Working and answer

Working and answer

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18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 15 marks)

18. Question 8 · part c · 8 marks · Working and answer

Working and answer

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19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

6 $f(x) = 4x^2 - 3x - 5$

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(a) form an equation, with integer coefficients, that has roots $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$ **(6)**

$g(x) = 4x^2 + px + q$ where p and q are constants

Given that the equation $g(x) = 0$ has roots $3\alpha + \beta$ and $\alpha + 3\beta$

(b) find the value of p and the value of q **(5)**

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19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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19. Question 6 · part b · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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19. Question 6 · part b · 5 marks

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Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 11 marks)

19. Question 6 · part b · 5 marks · Working and answer

Working and answer

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20. Question 8 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

8

$$f(x) = 3x^2 - x + 4$$

$$g(x) = x^2 - px + q$$

The roots of the quadratic equation $f(x) = 0$ are α and β

The roots of the quadratic equation $g(x) = 0$ are $\left(\alpha + \frac{1}{\alpha}\right)$ and $\left(\beta + \frac{1}{\beta}\right)$

Without solving the equation $f(x) = 0$

(a) show that $p = \frac{7}{12}$

(3)

(b) Find the value of q

(4)

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20. Question 8 · part a · 3 marks · Working and answer

Working and answer

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21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

8

$$f(x) = 3x^2 - x + 4$$

$$g(x) = x^2 - px + q$$

The roots of the quadratic equation $f(x) = 0$ are α and β

The roots of the quadratic equation $g(x) = 0$ are $\left(\alpha + \frac{1}{\alpha}\right)$ and $\left(\beta + \frac{1}{\beta}\right)$

Without solving the equation $f(x) = 0$

(a) show that $p = \frac{7}{12}$

(3)

(b) Find the value of q

(4)

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21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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21. Question 8 · part b · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 7 marks)

21. Question 8 · part b · 4 marks · Working and answer

Working and answer

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22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

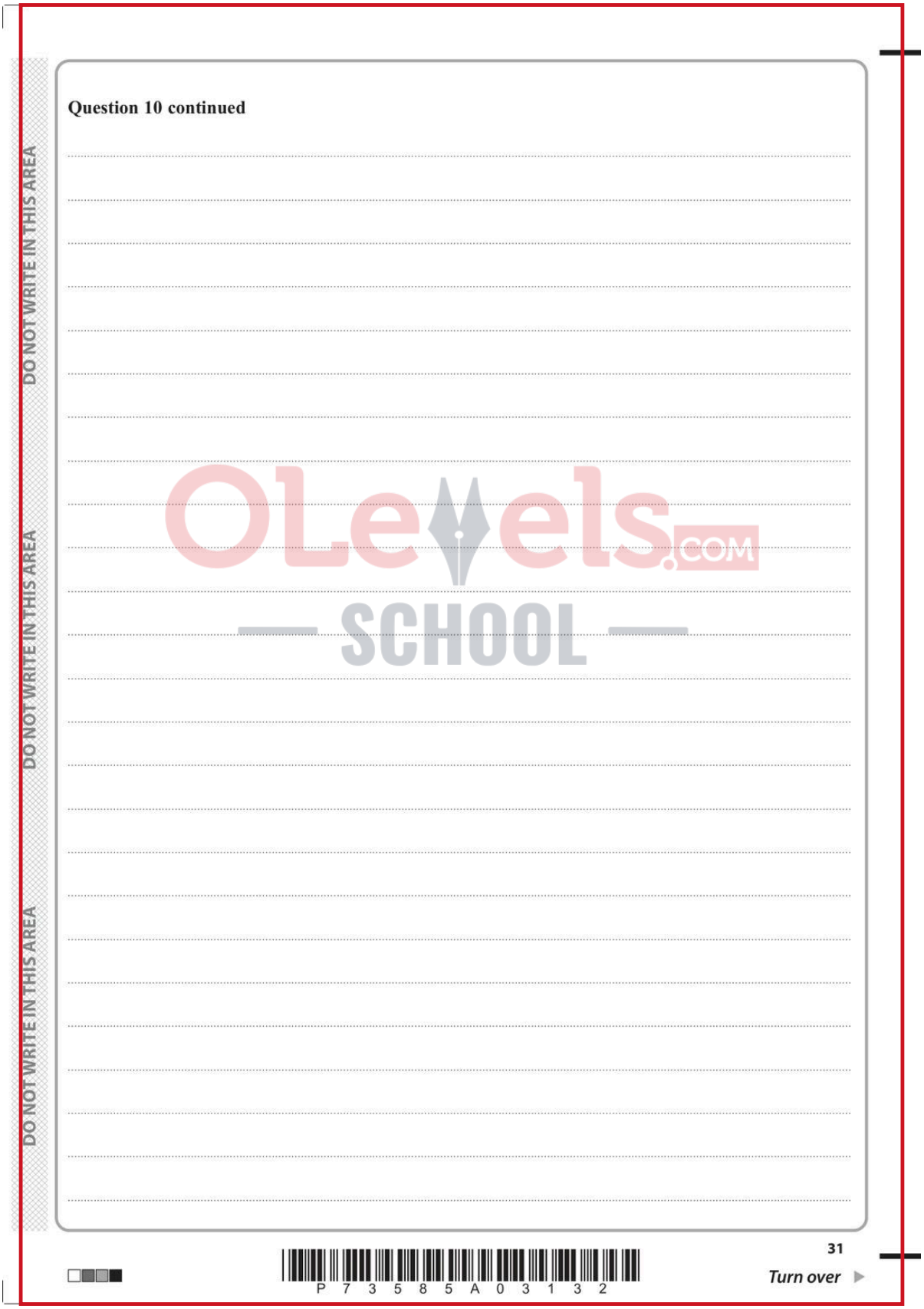
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22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.



Question 10 continued

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22. Question 10 · part e · 6 marks

2024 NOV · Paper 01

Further practice · Attempt part e only.

Question 10 continued

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(Total for Question 10 is 18 marks)

22. Question 10 · part e · 6 marks · Working and answer

Working and answer

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